

TOTAL: 19/20

COMMUNICATION:

Given $f(x) = \frac{3x}{x^2 - 1}$, evaluate each of the following limits.

a) $\lim_{x \rightarrow -1^+} f(x)$

$f(-0.9999) = \frac{3(-0.9999)}{(-0.9999)^2 - 1}$
 $= 14999.25$

(3) $\lim_{x \rightarrow -1^+} f(x) = \infty$

b) $\lim_{x \rightarrow 1^-} f(x)$

$f(0.9999) = \frac{3(0.9999)}{(0.9999)^2 - 1}$
 $= -14999.25$

$\lim_{x \rightarrow 1^-} f(x) = -\infty$

c) $\lim_{x \rightarrow \pm\infty} f(x)$

$= \lim_{x \rightarrow \pm\infty} \frac{x(3)}{x^2(1 - \frac{1}{x^2})}$
 $= \lim_{x \rightarrow \pm\infty} \frac{3}{x}$

$\lim_{x \rightarrow \infty} f(x) = 0^+$

$\lim_{x \rightarrow -\infty} f(x) = 0^-$

2. Given the graph of $f(x)$, sketch a possible graph for $f'(x)$. Justify your sketch.

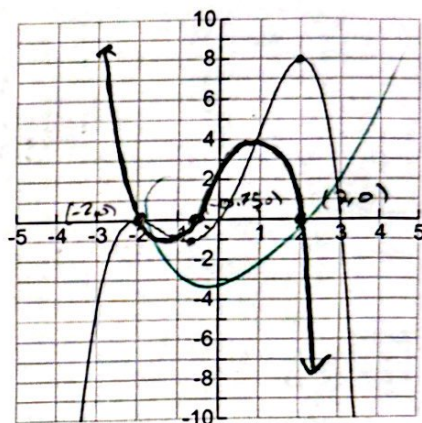
There appear to be local extrema at $x = -2, -0.75, 2$ roughly.

$f(x)$ is increasing from: $(-\infty, -2) \cup (-0.75, 2)$

$f(x)$ is decreasing from: $(-2, -0.75) \cup (2, \infty)$

Therefore: $f'(x)$ will have:

- Zeros at $x = -2, -0.75, 2$
- $f'(x) > 0$ on intervals $(-\infty, -2) \cup (-0.75, 2)$
- $f'(x) < 0$ on intervals $(-2, -0.75) \cup (2, \infty)$



KNOWLEDGE:

1. The derivative of a function $g(x)$ is $g'(x) = (2x+1)^6(x+4)^3$.

Determine on what interval $g(x)$ is decreasing.

Crit points: $x = -\frac{1}{2}, -4$

	$(-\infty, -4)$	$(-4, -\frac{1}{2})$	$(-\frac{1}{2}, \infty)$
$g'(x)$	-	+	+
$g(x)$	↘	↗	↗

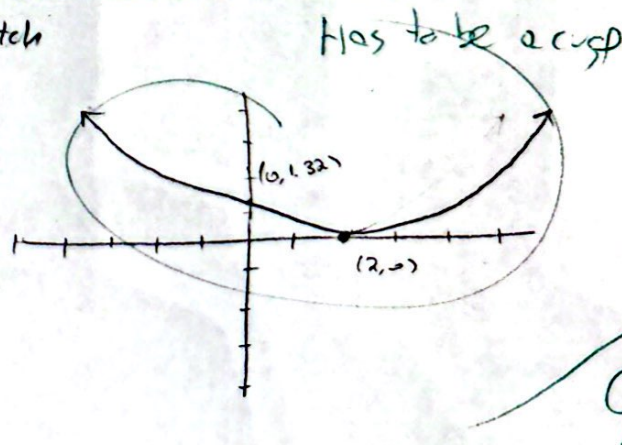
Therefore $g(x)$ is decreasing on the interval $x < -4$.

2. Use the First Derivative Test to determine intervals of increase/decrease, and to classify any local extrema if $f(x) = (x-2)^{\frac{2}{3}}$. Provide rough sketch

$f'(x) = \frac{2}{3}(x-2)^{-\frac{1}{3}}$

$= \frac{2}{5\sqrt[3]{x-2}}$ → Crit points $x = 2$

	$(-\infty, 2)$	$(2, \infty)$
$f'(x)$	-	+
$f(x)$	↘	↗
	local min	
	$(2, 0)$	



2. Sketch the graph of $f(x) = \frac{x^2 + 4x + 4}{x + 3}$ by considering behaviour around the applicable asymptote.

Include the first derivative test to classify local extrema.

(6)

$$y = \text{int}$$

$$y = \frac{4}{3}$$

$$x = -2$$

V.A.

$$x = -3$$

$$\lim_{x \rightarrow -3^-} f(x) = -\infty$$

$$\lim_{x \rightarrow -3^+} f(x) = \infty$$

H.A.

$$\lim_{x \rightarrow \infty} \frac{x^2(1 + \frac{4}{x} + \frac{4}{x^2})}{x(1 + \frac{3}{x})}$$

$$\lim_{x \rightarrow \infty} x = \pm \infty$$

No H.A.

$$f(x) = \frac{(x+2)^2}{(x+3)}$$

Obligate asymptote

$$x+3 \sqrt{x^2+4x+4}$$

$$-(x^2+3x)$$

$$x+4$$

$$\frac{x+4}{x+3}$$

doesn't cross

$$y = x + 1$$

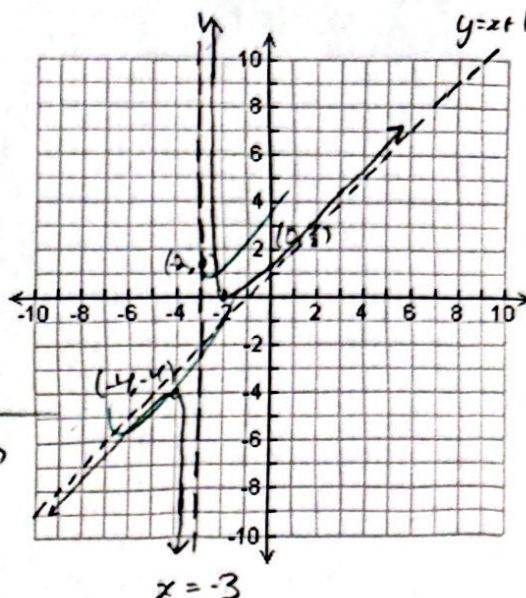
$$f'(x) = \frac{(2x+4)(x+3) - (x^2+4x+4)(1)}{(x+3)^2}$$

$$= \frac{2x^2 + 10x + 12 - x^2 - 4x - 4}{(x+3)^2}$$

$$= \frac{x^2 + 6x + 8}{(x+3)^2}$$

$$f'(x) = \frac{(x+4)(x+2)}{(x+3)^2}$$

$$x = -4, -2, -3$$



$(-\infty, -4)$	$(-4, -3)$	$(-3, -2)$	$(-2, \infty)$
$f'(x)$	+	-	-
$f(x)$	→	→	→
	Smooth local max	V.A.	Smooth local min
	$(-4, -4)$		$(-2, 0)$

APPLICATION:

1. Determine the values of a, b, c, and d, given $g(x) = ax^3 + bx^2 + cx + d$ has a local minimum at $(-1, 1)$ and a local maximum at $(0, 5)$.

$$g'(-1) = 0$$

$$g'(0) = 0$$

$$P(-1, 1)$$

$$P(0, 5)$$

$$5 = d$$

$$1 = a(-1)^3 + b(-1)^2 + c(-1) + 5$$

$$1 = -a + b - c + 5$$

$$c = 4 + b - a$$

$$g'(x) = 3ax^2 + 2bx + c$$

$$0 = c$$

$$0 = 3a(-1)^2 + 2b(-1)$$

$$0 = 3a - 2b$$

$$2b = 3a$$

$$b = \frac{3}{2}a$$

$$(-\infty, -1) \quad (-1, 0) \quad (0, \infty)$$

$f'(x)$	-	+	-
$f(x)$	→	→	→

$$a = 4 + \frac{3}{2}a$$

$$-\frac{1}{2}a = 4$$

$$a = -8$$

$$b = \frac{3}{2}(-8)$$

$$b = -12$$

$$c = 0$$

$$d = 5$$

$$g(x) = -8x^3 - 12x^2 + 5$$

Check

$$g'(x) = -24x^2 - 24x$$

$$= -24x(x+1)$$

$$\text{crit } x = 0, -1$$

$$(-\infty, -1) \quad (-1, 0) \quad (0, \infty)$$

$$g'(x)$$

$$- \quad + \quad -$$

$$\rightarrow \quad \rightarrow \quad \rightarrow$$

10